

Untitled

Changze Wu_1008837840

2024-03-01

#Q1

```
data = read.csv(file = "https://ckan0.cf.opendata.inter.prod-toronto.ca/dataset/21c83b32-d5a8-4106-a54f-010dbe49f6f2/resource/55d58477-50f5-490c-8da8-5953e3b26ca4/download/daily-shelter-overnight-service-occupancy-capacity-2022.csv")
```

```
pop = na.omit(data$CAPACITY_ACTUAL_ROOM)
```

```
#data %>% select(CAPACITY_ACTUAL_ROOM) -> p
#p = na.omit(p)
pop_avg = mean(pop)
pop_avg
```

```
## [1] 69.45697
```

```
x = sample(pop, size = 500)
qnorm(c(0.1, 0.9), mean=mean(x), sd=sd(x)/sqrt(500))
```

```
## [1] 63.99676 71.07924
```

```
t.test(x, conf.level = 0.8)
```

```
##
## One Sample t-test
##
## data: x
## t = 24.442, df = 499, p-value < 2.2e-16
## alternative hypothesis: true mean is not equal to 0
## 80 percent confidence interval:
## 63.99207 71.08393
## sample estimates:
## mean of x
## 67.538
```

#Q2

```
bootstrap = function() {  
  s = sample(x, size = 500, replace = T)  
  return (mean(s))  
}  
bootstrap_x_bar = replicate(100000, bootstrap())  
quantile(bootstrap_x_bar, c(0.1, 0.9))
```

```
##      10%      90%  
## 64.024 71.102
```